

GEORGE SAJU

Analyst | Zifo Technologies Pvt.Ltd

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PROFESSIONAL SUMMARY

Results-driven Data Scientist with over three years of experience in Python programming, data analysis, and API development. Expertise in utilizing advanced tools such as Pandas, Polars, and NumPy for numerical computing, complemented by proficiency in Matplotlib and Seaborn for impactful data visualization. Proven track record in building and integrating REST APIs using FastAPI, along with a solid foundation in parser development for scientific instruments like NMR, HPLC, and GC-MS to enhance data extraction and automate workflows. Committed to leveraging big data technologies such as Apache PySpark to efficiently process large datasets and facilitate data-informed decision-making through scalable analytical solutions. Retrieval-Augmented Generation (RAG) pipelines for intelligent information retrieval, document processing, and AI-driven applications. Proficient in integrating LLM-based solutions into real-world systems to enhance automation and decision-making. Strong foundation in big data technologies such as Apache Spark, enabling efficient processing of large-scale datasets and delivering scalable, data-driven solutions.

CORE COMPETENCIES AND SKILLS

- Programming: Python, SQL
- Data Analysis & Processing: Pandas, Polars, NumPy
- Machine Learning: Scikit-learn, PyTorch
- Data Visualization: Matplotlib, Seaborn
- Big Data Technologies: PySpark,
- Databases: MySQL, PostgreSQL, MongoDB
- API Development: FastAPI, REST APIs
- Data Engineering: ETL Pipelines, Data Cleaning, Data Transformation
- Scientific Data Processing: NMR, HPLC, GC-MS Data Parsing
- LLM : GPT, RAG
- Other Tools: Git,

PROFESSIONAL EXPERIENCE

Data Scientist | Analyst

Zifo Technologies Pvt. Ltd.

Experience: 3 Years

Duration: January 2023 – April 2026

- Designed and developed high-performance **data pipelines using Pandas, Polars, and NumPy**, achieving a **40% reduction in processing time** through optimized transformations and efficient data handling.
- Built and maintained **custom parsers for scientific instruments** including **NMR, HPLC, and GC-MS**, automating data extraction workflows and reducing manual effort by **80% while improving accuracy and data integrity**.
- Developed and optimized **RESTful APIs using FastAPI** for instrument and ELN integrations, delivering **60% faster data retrieval** compared to legacy Flask-based implementations.
- Created **interactive dashboards and analytical reports** using **Matplotlib, Seaborn** enabling stakeholders to identify trends and support data-driven decision-making, contributing to a **20% improvement in operational efficiency**.
- Implemented **machine learning models and predictive analytics** for scientific data processing, applying effective **feature engineering techniques** to enhance model performance.
- Collaborated closely with **data engineers, chemists, and business stakeholders** to deliver domain-specific analytical solutions aligned with research and business objectives.
- Developed Retrieval-Augmented Generation (RAG) pipelines by integrating vector-based search with LLMs to enable context-aware querying and intelligent knowledge retrieval from structured and unstructured data sources.

ACHIEVEMENTS

- Designed and implemented instrument parsers for NMR, HPLC, and GC-MS, reducing manual data extraction efforts by 80% and increasing accuracy.

- Built REST APIs using FastAPI, improving data retrieval speeds by 60% compared to traditional Flask-based APIs.
- Created interactive dashboards with Matplotlib and Seaborn that helped stakeholders identify trends and make data-driven decisions, leading to a 20% improvement in operational efficiency.
- Automated data extraction, transformation, and loading (ETL) processes.

PROJECTS

Generic Parser Development for Proprietary Formats, Designed and implemented parsers for multiple proprietary file formats (.spc, .D, .qgd, etc.) to automate data extraction, ensuring high accuracy and compatibility across scientific instruments. Intelligent Agent Integration with ELN Systems, Built REST API-based agents using FastAPI to seamlessly integrate Electronic Lab Notebooks (ELN) with instruments. Enabled automated data logging, reducing manual data entry efforts by 75% and improving traceability. Scientific Data Extraction & Predictive Analytics Pipeline, Developed end-to-end data pipelines to extract and preprocess analytical data from instruments such as NMR, GC-MS, and HPLC. Implemented ML-based predictive models to detect trends and optimize instrument performance, achieving 35% faster analysis cycles.

EDUCATION

- **Master of Science (M.Sc.) in Chemistry**
Mahatma Gandhi University, Kerala | 2022
- **Bachelor of Science (B.Sc.) in Chemistry**
Mahatma Gandhi University, Kerala | 2020

CERTIFICATIONS & LICENSE

- **Certified Data Scientist** – Illinois University
- **Scientific Data Analysis & Parser Development** – Zifo
- **Time Management & Productivity Training** – Zifo
- **Leadership & People Management Training** – Zifo
- **Project Management & Agile Training** – Zifo

TRAINING AND PROFESSIONAL DEVELOPMENT

- **Project Management & Agile Training:** Expertise in planning, execution, and monitoring of projects using Agile methodologies.
- **Leadership & Soft Skills Development:** Trained in people management, leadership, and effective communication to enhance collaboration.
- **Time Management & Productivity Training:** Learned advanced techniques for task prioritization, delegation, and efficiency improvement.
- **Technical & Scientific Data Processing:** Specialized training in scientific data analysis, parser development, and data integrity for research and pharmaceutical applications.
- **Client & Stakeholder Communication Training:** Developed strong skills in stakeholder management, negotiation, and client communication.

REFERENCES

- Available Upon Request